

Adaptive Load-Carrying Control using Quadrotors in a Tandem Configuration

Alexandre S. Brandão¹, Leonardo A. Fagundes-Júnior¹ and Pedro Castillo²

Abstract—This paper presents an adaptive control strategy for cooperative cargo transportation using quadcopters in a tandem configuration. By modeling the payload and the unmanned aerial vehicles (UAVs) as a unified rigid-body system, the proposed framework addresses the dynamic interactions among them, ensuring robust performance in tasks such as cargo transportation. The system uses an underactuated adaptive control approach, capable of dealing with variations in payload weight while maintaining stability and agility during flight. The proposed strategy is validated through numerical experiments, demonstrating its effectiveness in trajectory tracking tasks. The results show the system's ability to adapt the parameters of the system modeling to the observed and measured values, guaranteeing the tracking of the proposed trajectory and the robust payload handling. This work contributes to the development of cooperative aerial cargo transportation systems, with applications in transportation missions that exceed the individual capacity of each UAV.

I. INTRODUCTION

Unmanned aerial vehicles (UAVs), especially quadcopters, have emerged as tools for a wide range of applications, including search and rescue missions, area surveillance, mapping challenging terrain and inspecting infrastructure. Recently, the potential of UAVs for cooperative load transportation has gained significant interest. This involves deploying quadcopters to collaboratively grab, handle and transport payloads, and there are two main strategies: tethering the payload to the vehicle body or attaching the payload directly to the UAV [1]. The tethering approach, while introducing additional unactuated degrees of freedom, maintains the vehicle's maneuverability. In contrast, direct attachment simplifies the mathematical modeling of the system, but increases inertia, reducing the vehicle's agility.

From the perspective of aerial robotics, UAVs are intrinsically unstable, so their use for cargo transportation further increases the complexity of challenges from an engineering and control perspective. In response, several approaches have been proposed by the scientific community to present solutions to this navigation problem. Some of them include: trajectory optimization to suppress cargo swing [2]–[4], computer vision-based navigation techniques to overcome the need for motion capture systems or autonomous flight only in indoor environments [5], aggressive maneuvers to optimize energy consumption and deal with challenging navigation

environments [6], [7], aerial cargo manipulation using claws, hands, and skilled robotic arms [7]–[9], UAV interaction with other types of robots [10]–[12], as well as various approaches to effectively transport cargo using teams of UAVs [13], [14].

A suspended cable attached to a UAV significantly affects its flight dynamics, making stabilization and control more challenging. This issue is especially pronounced when the cargo's mass, size, shape, or cable parameters are unknown. As a result, various control strategies focus on minimizing or dampening the impact of the cargo by treating it as a disturbance. These approaches often incorporate adaptive and/or robust techniques [15]–[17], or estimate the oscillation angle of the cargo to counteract its effects [18]. In a similar vein, artificial intelligence methods are employed to estimate the weight of unknown cargo and mitigate parametric and non-parametric uncertainties in flight using neural network-based models [19]–[21], as well as to update PID controller gains with the use of Wavelet Neural Networks [22].

This paper focuses on cooperative cargo transportation using quadcopters in a tandem configuration [23], where two UAVs work together to transport a shared payload, as shown in Figure 1. This approach is particularly advantageous for tasks that require the transportation of payloads that exceed the thrust capacity of a single UAV, offering benefits such as greater payload capacity, system redundancy, and improved safety. However, this strategy presents complex challenges, including the need for precise coordination between the agents and the management of dynamic effects such as payload oscillations and mutual aerodynamic interactions between the quadcopters.

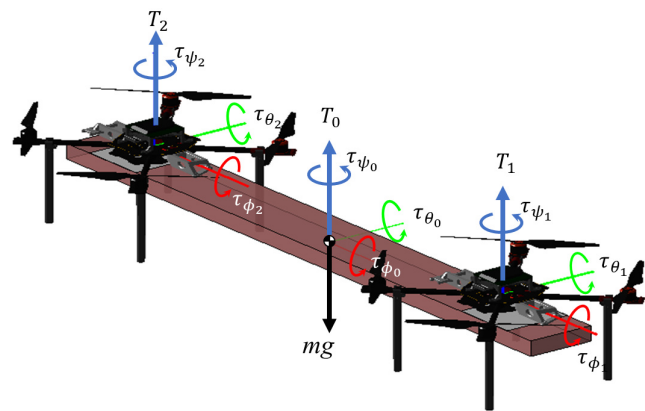


Fig. 1. Tandem configuration using two quadrotors.

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¹A.S. Brandão and L.A. Fagundes-Jr are with the Nucleus of Specialization in Robotics, Department of Electrical Engineering, Universidade Federal de Viçosa, 36570-900 Viçosa, MG, Brazil { alexandre.brandao, leonardo.fagundes}@ufv.br

²P. Castillo is with Université de Technologie de Compiègne, CNRS, Heudiasyc, 60203 Compiègne Cedex, France. castillo@hds.uto.fr.

In such a context, payload oscillations refer to the undesired swinging or vibratory motion of the shared load, which can arise when the movement of one UAV unintentionally pushes or pulls the other via the coupled payload. This interaction can introduce a coupling force between the agents that, if not properly damped, may lead the system into a resonant behavior. In such cases, the amplitude of oscillation of the UAV-load assembly may increase progressively, potentially destabilizing the formation and leading to the collapse of the navigation strategy.

Although previous research has shown promising results in single-agent and multi-agent cargo transportation systems, several critical challenges have yet to be addressed. These include stabilizing pendulum-like dynamics in load suspended from cables, developing robust mathematical models that take into account the various flight modes, and designing controllers capable of mitigating dynamic disturbances such as wind gusts and payload weight variations during flight. In cooperative systems, an additional layer of complexity arises from the need to ensure that each agent can accurately track a desired trajectory while compensating for the forces exerted by the payload and other UAVs [9].

Similar to our previous work [16], which focused on cooperative load transportation with two quadrotors using a cable-suspended approach, this paper introduces an adaptive control strategy for quadrotors operating in a tandem configuration for collaborative payload transport. By modeling both the payload and UAVs as a unified rigid-body system, we develop a robust control framework capable of addressing the dynamic interactions between the UAVs and the payload. This framework ensures precise and stable load transportation, even under external disturbances and variations in payload characteristics. Given this context, this paper proposes an adaptive controller that will enable the UAVs to adapt to such situations. Thus, our main contributions are *a)* the proposal of an adaptive controller for bar-shaped payload transportation; *b)* the stability analysis of the proposed controller demonstrated in the sense of Lyapunov; *c)* an online mass and moments estimation that works for grasping and suspended load transportation scenarios.

The paper is organized as follows: Section II explores the control of a rigid-body system in near-hovering conditions, covering the underactuated dynamic model, parametric representation, adaptive control design, and a stability analysis to validate the approach. Section III extends the discussion to a tandem-copter configuration, outlining the control strategy for enabling multiple UAVs to cooperate effectively in transporting a shared payload. Section IV presents the results of numerical simulations, offering a detailed analysis of their practical implications. Finally, Section V concludes the paper by summarizing the key contributions and proposing directions for future research.

II. CONTROLLING A RIGID-BODY IN NEAR-HOVERING

This section presents the Euler-Lagrange formulation for a generic rigid body moving in 3-D space, similar to the approach in our previous work [24]. In this paper, this

modeling is applied to represent both the tandem system and the quadrotors that compose it.

To introduce the modeling, we first define the UAV pose in Cartesian space, represented in generalized coordinates as $\mathbf{q} = [\xi^\top \ \eta^\top]^\top$, where $\xi = [x \ y \ z]^\top$ denotes the longitudinal, lateral, and normal position/displacement in the inertial frame, and $\eta = [\phi \ \theta \ \psi]^\top$ corresponds to the roll, pitch, and yaw Tait-Bryan angles.

Considering that most cargo transport tasks are performed at low speeds without aggressive maneuvers, we assume the UAVs fly in near-hovering conditions. Mathematically, this condition holds when the pitch and roll angles are small, typically less than 5° in magnitude. In this scenario, we can approximate $s_\star \approx \star$ and $c_\star \approx 1$, where $s_\star = \sin \star$ and $c_\star = \cos \star$.

Now writing the Lagrangian function that represents the difference between the total kinematic and total potential energies, we have

$$L = \frac{1}{2} m \dot{\xi}^\top \dot{\xi} + \frac{1}{2} \boldsymbol{\Omega}^\top \mathbf{I} \boldsymbol{\Omega} - mgz, \quad (1)$$

where m is the total mass, g is the gravitational acceleration, $\mathbf{I} = \text{diag}\{I_x, I_y, I_z\}$ is the diagonal matrix of the moments-of-inertia, and $\boldsymbol{\Omega}$ represents the angular velocities in the body frame, which is given by

$$\boldsymbol{\Omega} = \begin{bmatrix} 1 & 0 & -s_\theta \\ 0 & c_\phi & s_\phi c_\theta \\ 0 & -s_\phi & c_\phi c_\theta \end{bmatrix} \dot{\boldsymbol{\eta}} \approx \begin{bmatrix} 1 & 0 & -\theta \\ 0 & 1 & \phi \\ 0 & -\phi & 1 \end{bmatrix} \dot{\boldsymbol{\eta}}. \quad (2)$$

Remark 1: Properties of skew-symmetric matrices can be used to represent cross products, i.e., $\boldsymbol{\Omega} \times \mathbf{I} \boldsymbol{\Omega} = S(\boldsymbol{\Omega}) \mathbf{I} \boldsymbol{\Omega}$.

Taking $L_R = \frac{1}{2} \boldsymbol{\Omega}^\top \mathbf{I} \boldsymbol{\Omega}$ as the rotational Lagrangian, after some manipulations, we get

$$L_R = \frac{1}{2} \left[I_x (\dot{\phi}^2 - 2\theta\dot{\phi}\dot{\psi} + \theta^2\dot{\psi}^2) + I_y (\dot{\theta}^2 + 2\phi\dot{\theta}\dot{\psi} + \phi^2\dot{\psi}^2) + I_z (\dot{\psi}^2 - 2\phi\dot{\theta}\dot{\psi} + \phi^2\dot{\theta}^2) \right]. \quad (3)$$

Now applying the Euler-Lagrange constraint

$$\frac{d}{dt} \frac{\partial L_R}{\partial \dot{\boldsymbol{\eta}}} - \frac{\partial L_R}{\partial \boldsymbol{\eta}} = \boldsymbol{\tau},$$

we obtain

$$\begin{aligned} \tau_\phi &= I_x (\ddot{\phi} - \theta\ddot{\psi} - \dot{\theta}\dot{\psi}) \\ &+ I_y (-\dot{\theta}\dot{\psi} - \phi\dot{\psi}^2) + I_z (\dot{\theta}\dot{\psi} - \phi\dot{\theta}^2) \end{aligned} \quad (4a)$$

$$\begin{aligned} \tau_\theta &= I_x (\dot{\phi}\dot{\psi} - \theta\dot{\psi}^2) \\ &+ I_y (\ddot{\theta} + \phi\ddot{\psi} + \dot{\phi}\dot{\psi}) + I_z (\phi^2\ddot{\theta} - \phi\ddot{\psi} - \dot{\phi}\dot{\psi} + 2\phi\dot{\phi}\dot{\theta}) \end{aligned} \quad (4b)$$

$$\begin{aligned} \tau_\psi &= I_x (-\theta\ddot{\phi} + \theta^2\ddot{\psi} - \dot{\phi}\dot{\theta} + 2\theta\dot{\theta}\dot{\psi}) \\ &+ I_y (\phi\ddot{\theta} + \phi^2\ddot{\psi} + \dot{\phi}\dot{\theta} + 2\phi\dot{\phi}\dot{\psi}) + I_z (-\phi\ddot{\theta} + \ddot{\psi} + \dot{\phi}\dot{\theta}) \end{aligned} \quad (4c)$$

Next, we can reorganize these equations and write the rotational dynamic model as

$$\mathbf{M}_R \ddot{\boldsymbol{\eta}} + \mathbf{C}_R \dot{\boldsymbol{\eta}} = \boldsymbol{\tau}, \quad (5)$$

with the inertial matrix given by

$$\mathbf{M}_R = \begin{bmatrix} I_x & 0 & -I_x \theta \\ 0 & I_y + I_z \phi^2 & -I_z \phi + I_y \phi \\ -I_x \theta & I_y \phi - I_z \phi & I_x \theta^2 + I_y \phi^2 + I_z \end{bmatrix}$$

and the Coriolis matrix expressed by its terms c_{ij} , after computing the Christoffel symbols,

$$\begin{aligned} c_{11} &= 0 \\ c_{12} &= -I_z \phi \dot{\theta} + \frac{1}{2} (-I_x - I_y + I_z) \dot{\psi} \\ c_{13} &= \frac{1}{2} (-I_x - I_y + I_z) \dot{\theta} - I_y \phi \dot{\psi} \\ c_{21} &= I_z \phi \dot{\theta} + \frac{1}{2} (I_x + I_y - I_z) \dot{\psi} \\ c_{22} &= I_z \phi \dot{\phi} \\ c_{23} &= \frac{1}{2} (I_x + I_y - I_z) \dot{\phi} - I_x \theta \dot{\psi} \\ c_{31} &= \frac{1}{2} (-I_x + I_y - I_z) \dot{\theta} + I_y \phi \dot{\psi} \\ c_{32} &= \frac{1}{2} (-I_x + I_y - I_z) \dot{\phi} + I_x \theta \dot{\psi} \\ c_{33} &= I_y \phi \dot{\phi} + I_x \theta \dot{\theta}. \end{aligned}$$

Now taking the remaining part of (1) and applying the Euler-Lagrange constraint once again, we get the translational dynamic model given by

$$\mathbf{M}_T \ddot{\boldsymbol{\xi}} + \mathbf{g} = \mathbf{f}, \quad (6)$$

with $\mathbf{M}_T = m\mathbb{I}_{3 \times 3}$, $\mathbf{g} = [0 \ 0 \ mg]^\top$ and $\mathbf{f}$ is the forces acting in the Cartesian axis.

Note that the rigid-body configuration uses the geometric distribution to compute the forces $\mathbf{f} = [f_x \ f_y \ f_z]^\top$ and torques $\boldsymbol{\tau} = [\tau_\phi \ \tau_\theta \ \tau_\psi]^\top$ acting on the system's body, which are finally used in the rigid-body model to determine the displacement in the Cartesian space.

Remark 2 (Abstract Inputs): The forces produced in the UAV body frame is $\mathbf{f} = [0 \ 0 \ T]^\top$, where $T = \sum_{i=1}^4 f_i$, with f_i being the thrust generated by each propeller. By the way, $\boldsymbol{\tau}$ is the torques acting on the UAV body frame.

Finally, we combine (5) and (6) to get the near-hovering model of a rigid-body as

$$\begin{bmatrix} \mathbf{M}_T & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_R \end{bmatrix} \begin{bmatrix} \ddot{\boldsymbol{\xi}} \\ \ddot{\boldsymbol{\eta}} \end{bmatrix} + \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_R \end{bmatrix} \begin{bmatrix} \dot{\boldsymbol{\xi}} \\ \dot{\boldsymbol{\eta}} \end{bmatrix} + \begin{bmatrix} \mathbf{g} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ \boldsymbol{\tau} \end{bmatrix} \quad (7)$$

or simply

$$\mathbf{M} \ddot{\mathbf{q}} + \mathbf{C} \dot{\mathbf{q}} + \mathbf{g} = \begin{bmatrix} \mathbf{f} \\ \boldsymbol{\tau} \end{bmatrix}. \quad (8)$$

Remark 3 (Dynamic model properties): Following this formulation we guarantee the properties: $\mathbf{M}$ is a positive defined symmetric matrix and $\mathbf{M} - 2\mathbf{C}$ is a non-zero skew-symmetric matrix.

A. Underactuated Dynamic Model

Underactuated systems are characterized by having fewer actuators than degrees of freedom (DoF). As a result, they cannot directly produce accelerations in each DoF at every specific instant of time without indirectly influencing the dynamics of other degrees of freedom.

Remark 4 (Passivity Concept): Here, a passive DoF is one whose dynamic is not directly actuated but instead influenced by the system's strong inertial coupling, meaning its dynamics are passively governed by the active variables.

In essence, the control of non-actuated variables in these systems is governed by the coupling between them and the actuated variables (directly associated to actuators). Examples of such systems include various flying robotic platforms, such as helicopters, airplanes, dirigibles and quadcopters, which have this underactuated nature.

Before expressing (8) as a underactuated system, let's take the control inputs of quadrotor UAV, whose propellers produce thrust on its z -axis direction (in its body frame). So we have

$$\mathbf{f} = \underbrace{\begin{bmatrix} c_\psi c_\theta & c_\psi s_\theta s_\phi - s_\psi c_\phi & c_\psi s_\theta c_\phi + s_\psi s_\phi \\ s_\psi c_\theta & s_\psi s_\theta s_\phi + c_\psi c_\phi & s_\psi s_\theta c_\phi - c_\psi s_\phi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{bmatrix}}_{\mathbf{R}} \begin{bmatrix} 0 \\ 0 \\ T \end{bmatrix}, \quad (9)$$

where $T = \sum_{i=1}^4 f_i$, being f_i the thrust produced by each i -th propeller, and $\mathbf{R} \in SO(3)$ is the rotational matrix considering the Roll-Pitch-Yaw Euler angles from the inertial frame to the vehicle body frame. Using (6) and replacing (9), we obtain

$$\mathbf{R}^\top \mathbf{M}_T \ddot{\boldsymbol{\xi}} + \mathbf{R}^\top \mathbf{g} = [0 \ 0 \ T]^\top, \quad (10)$$

which is clearly an underactuated system.

Now using (10) on (8), we get the dynamic model represented as a underactuated system

$$\begin{bmatrix} \mathbf{M}_{uu} & \mathbf{M}_{ua} \\ \mathbf{M}_{au} & \mathbf{M}_{aa} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{q}}_u \\ \ddot{\mathbf{q}}_a \end{bmatrix} + \begin{bmatrix} \mathbf{C}_{uu} & \mathbf{C}_{ua} \\ \mathbf{C}_{au} & \mathbf{C}_{aa} \end{bmatrix} \begin{bmatrix} \dot{\mathbf{q}}_u \\ \dot{\mathbf{q}}_a \end{bmatrix} + \begin{bmatrix} \mathbf{g}_u \\ \mathbf{g}_a \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \boldsymbol{\tau}_a \end{bmatrix}, \quad (11)$$

where $\boldsymbol{\tau}_a = [T \ \boldsymbol{\tau}]^\top$.

Notice that the subscripts u and a are used to indicate the unactuated and actuated variables, respectively. In this work the unactuated variables are $\mathbf{q}_u = [x \ y]^\top$, and the actuated ones are $\mathbf{q}_a = [z \ \phi \ \theta \ \psi]^\top$.

B. Parametric Representation

In the design of model-based control systems, the need for precise knowledge of the dynamics of the system and its mathematical description is common. However, it is not always possible to obtain such data with precision, hence some uncertainties appear in the model, which are classified as parametric or structural, in the literature. In the case of this work, the focus is on the parametric uncertainties, associated with the imprecise knowledge of the physical parameters, resulting from the lack of a priori information of their values or their alteration during the execution of a task, such as, lifting and transporting a cargo.

Observing the Lagrangian function (1), we conclude the system parameters are $\Theta = [m \ I_x \ I_y \ I_z]^\top$. In order to evidence the second-order constraint of this underactuated system, let's remove (11) from the matrix notation to obtain the actuated one

$$\mathbf{M}_{aa}\ddot{\mathbf{q}}_a + \underbrace{\mathbf{C}_{aa}\dot{\mathbf{q}}_a + \mathbf{g}_a + \mathbf{M}_{au}\ddot{\mathbf{q}}_u + \mathbf{C}_{au}\dot{\mathbf{q}}_u}_{\mathbf{E}_a} = \boldsymbol{\tau}_a, \quad (12)$$

after applying the approximation for small angles in $\mathbf{R}$ and performing some algebraic manipulations, we obtain

$$\boldsymbol{\tau}_a = \mathbf{Y}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})\boldsymbol{\Theta} \quad (13)$$

where $\mathbf{Y}(\star)$ is the regression matrix, whose terms are

$$\begin{aligned} Y_{11} &= (\phi s_\psi + \theta c_\psi)\ddot{x} + (\theta s_\psi - \phi c_\psi)\ddot{y} + \ddot{z} + g \\ Y_{22} &= \ddot{\phi} - \theta\ddot{\psi} - \dot{\theta}\dot{\psi} \\ Y_{23} &= -\dot{\theta}\dot{\psi} - \phi\dot{\psi}^2 \\ Y_{24} &= \dot{\theta}\dot{\psi} - \phi\dot{\theta}^2 \\ Y_{32} &= \dot{\phi}\dot{\psi} - \theta\dot{\psi}^2 \\ Y_{33} &= \ddot{\theta} + \phi\ddot{\psi} + \dot{\phi}\dot{\psi} \\ Y_{34} &= \phi^2\ddot{\theta} - \phi\ddot{\psi} - \dot{\phi}\dot{\psi} + 2\phi\dot{\phi}\dot{\theta} \\ Y_{42} &= \theta^2\ddot{\psi} - \theta\ddot{\phi} - \dot{\phi}\dot{\theta} + 2\theta\dot{\theta}\dot{\psi} \\ Y_{43} &= \phi\ddot{\theta} + \phi^2\ddot{\psi} + \dot{\phi}\dot{\theta} + 2\phi\dot{\phi}\dot{\psi} \\ Y_{44} &= \ddot{\psi} - \phi\ddot{\theta} - \dot{\phi}\dot{\theta} \\ Y_{12} &= Y_{13} = Y_{14} = Y_{21} = Y_{31} = Y_{41} = 0. \end{aligned}$$

Note that the previous formulation applies only to small roll and pitch angles, also known as near-hovering.

The representation of the underactuated dynamic model of a rigid body in parameterized form is one of the contributions of this manuscript, once researchers interested in implementing adaptive control strategies, such as the one presented in the sequel, can directly use the regression matrix (13).

C. Adaptive Underactuated Control

To show the relevance of this section in the context of a load-carrying task, let's consider that the quadrotors encharge to transport an object do not have any prior information about the object's mass and inertia tensor. Thus, only after grasping it, the quadrotors' controller is able to sense and observe this new situation, and then starts reacting to compensate the dynamic changes. It is worth mentioning that the dimensions and shape of this objective can be obtained by information from an aircraft edge vision system, however, this is beyond the scope of this work.

In order to do a partial feedback linearization in (12), we propose the control law

$$\boldsymbol{\tau}_a = \hat{\mathbf{M}}_{aa}\boldsymbol{\nu}_a + \hat{\mathbf{E}}_a, \quad (14)$$

with

$$\boldsymbol{\nu}_a = \ddot{\mathbf{q}}_{ad} + \mathbf{K}_v\dot{\tilde{\mathbf{q}}}_a + \mathbf{K}_p\tilde{\mathbf{q}}_a. \quad (15)$$

Notice that the subscript d indicates the desired value of some variable, and $\tilde{\mathbf{q}}_a = \mathbf{q}_{ad} - \mathbf{q}_a$ represents the control error, while $\mathbf{K}_v, \mathbf{K}_p \in \mathbb{R}^{4 \times 4}$ are positive definite diagonal

gain matrices. In addition, $[\hat{\star}]$ indicates that the matrices and vectors are given in terms of estimated parameters, i.e., it contemplates the modeling uncertainties.

The system's closed-loop representation can be written as

$$\hat{\mathbf{M}}_{aa}\boldsymbol{\nu}_a = \mathbf{M}_{aa}\ddot{\mathbf{q}}_a + \mathbf{E}_a - \hat{\mathbf{E}}_a. \quad (16)$$

Once the parametric estimate is not exact, its error can be expressed by $[\tilde{\star}] = [\star] - [\hat{\star}]$, and we can express $\tilde{\mathbf{M}}_{aa} = \mathbf{M}_{aa} - \hat{\mathbf{M}}_{aa}$, $\tilde{\mathbf{E}}_a = \mathbf{E}_a - \hat{\mathbf{E}}_a$ and $\tilde{\boldsymbol{\Theta}} = \boldsymbol{\Theta} - \hat{\boldsymbol{\Theta}}$.

Finally, rearranging (16) and then using (15) and observing (12)-(13), the closed-loop system becomes

$$\begin{aligned} \hat{\mathbf{M}}_{aa}(\boldsymbol{\nu}_a - \ddot{\mathbf{q}}_a) &= \tilde{\mathbf{M}}_{aa}\ddot{\mathbf{q}}_a + \tilde{\mathbf{E}}_a \\ \ddot{\mathbf{q}}_a + \mathbf{K}_v\dot{\tilde{\mathbf{q}}}_a + \mathbf{K}_p\tilde{\mathbf{q}}_a &= \hat{\mathbf{M}}_{aa}^{-1}\mathbf{Y}\tilde{\boldsymbol{\Theta}}. \end{aligned} \quad (17)$$

D. Stability Demonstration

The stability analysis is demonstrated in the sense of Lyapunov for the closed-loop system (17). Let's take the positive radially-unbounded candidate function

$$V(\tilde{\mathbf{q}}_a, \dot{\tilde{\mathbf{q}}}_a, \tilde{\boldsymbol{\Theta}}) = \frac{1}{2}\tilde{\mathbf{q}}_a^\top \mathbf{K}_p \tilde{\mathbf{q}}_a + \frac{1}{2}\dot{\tilde{\mathbf{q}}}_a^\top \dot{\tilde{\mathbf{q}}}_a + \frac{1}{2}\tilde{\boldsymbol{\Theta}}^\top \mathbf{Q} \tilde{\boldsymbol{\Theta}}, \quad (18)$$

computing its first time derivative and then replacing (17), we get

$$\begin{aligned} \dot{V}(\star) &= \tilde{\mathbf{q}}_a^\top \mathbf{K}_p \dot{\tilde{\mathbf{q}}}_a + \dot{\tilde{\mathbf{q}}}_a^\top \ddot{\tilde{\mathbf{q}}}_a + \tilde{\boldsymbol{\Theta}}^\top \mathbf{Q} \dot{\tilde{\boldsymbol{\Theta}}} \\ &= \dot{\tilde{\mathbf{q}}}_a^\top (\hat{\mathbf{M}}_{aa}^{-1}\mathbf{Y}\tilde{\boldsymbol{\Theta}} - \mathbf{K}_v\dot{\tilde{\mathbf{q}}}_a) + \tilde{\boldsymbol{\Theta}}^\top \mathbf{Q} \dot{\tilde{\boldsymbol{\Theta}}}. \end{aligned} \quad (19)$$

Assuming the adaptation law as

$$\dot{\tilde{\boldsymbol{\Theta}}} = \mathbf{Q}^{-1}(\hat{\mathbf{M}}_{aa}^{-1}\mathbf{Y})^\top \dot{\tilde{\mathbf{q}}}_a, \quad (20)$$

and also considering the system parameters do not vary during the phases of task execution, we have $\dot{\tilde{\boldsymbol{\Theta}}} = -\dot{\hat{\boldsymbol{\Theta}}}$, as $\boldsymbol{\Theta}$ is constant. This last assumption is pertinent because a carry-transport task consists of a phase without the cargo and the another with it. First, the quadrotor is navigating under the action of its own weight, without requiring any changes to its parameters. Later, after grasping the load and starting lifting and transporting stage, its dynamics change, and thus an adaptation process runs to update the parameters. Finally, when the drone releases the cargo, it returns to the first phase, thus requiring a new parametric adaptation.

Replacing (20) in (19), we get

$$\dot{V}(\tilde{\mathbf{q}}_a, \dot{\tilde{\mathbf{q}}}_a, \tilde{\boldsymbol{\Theta}}) = -\dot{\tilde{\mathbf{q}}}_a^\top \mathbf{K}_v \dot{\tilde{\mathbf{q}}}_a \leq 0. \quad (21)$$

As a direct conclusion, we have that the variables involved in the Lyapunov function are bounded, i.e., $\tilde{\mathbf{q}}_a, \dot{\tilde{\mathbf{q}}}_a, \tilde{\boldsymbol{\Theta}} \in L_\infty$.

Now taking the time derivative of (21) and replacing (17), we get

$$\ddot{V}(\star) = -2\dot{\tilde{\mathbf{q}}}_a^\top \mathbf{K}_v (\hat{\mathbf{M}}_{aa}^{-1}\mathbf{Y}\tilde{\boldsymbol{\Theta}} - \mathbf{K}_v\dot{\tilde{\mathbf{q}}}_a). \quad (22)$$

As a consequence of (21), we can guarantee that $\dot{V}$ is bounded, once all variables on it are bounded. Therefore, we conclude using Barbalat's lemma that $\tilde{\mathbf{q}}_a, \dot{\tilde{\mathbf{q}}}_a \in L_2$ and so $\tilde{\mathbf{q}}_a, \dot{\tilde{\mathbf{q}}}_a \rightarrow \mathbf{0}$ to $t \rightarrow \infty$. In other words, we have that the actuated variables converge asymptotically, while the estimate parameters remain ultimately bounded.

III. CONTROLLING A TANDEM-COPTER CONFIGURATION

This section aims to control a rigid body in a configuration similar to a tandem helicopter, as can be seen in Figure 1. In this work, each rotor responsible for generating the propulsion will be represented with a quadrotor. Therefore, this new rigid body has two UAVs as actuators at its ends.

Having in mind that the drone in question is not capable of generating longitudinal or lateral propulsion, at its own reference system, the proposed tandem configuration will not be either. Looking at Figure 2 and taking the distance of each drone to the center of gravity (point of control) of the tandem configuration, we have the following relationship between the control signals

$$\begin{bmatrix} T_0 \\ \tau_{\phi_0} \\ \tau_{\theta_0} \\ \tau_{\psi_0} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ -l_1 & 0 & 1 & 0 & l_2 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} T_1 \\ \tau_{\phi_1} \\ \tau_{\theta_1} \\ T_2 \\ \tau_{\phi_2} \\ \tau_{\theta_2} \\ \tau_{\psi_2} \end{bmatrix}, \quad (23)$$

where the subscript 0 represents the tandem, and the 1 and 2 indicate each drone in the formation. Note that, the

- total vertical force (T_0) is the result of the sum of the vertical propulsion forces of the drones:

$$T_0 = T_1 + T_2.$$

- torque around the x -axis (τ_{ϕ_0}) is the result of the sum of the rotational torques around the x -axis generated by each drone:

$$\tau_{\phi_0} = \tau_{\phi_1} + \tau_{\phi_2}.$$

- torque around the y -axis (τ_{θ_0}) is the result of the sum of the rotational torques around the y -axis generated by each drone, in addition to the torque produced by the propulsion force of each UAV at the center of the bar, considering the distances l_i from the coupling point:

$$\tau_{\theta_0} = -l_1 T_1 + \tau_{\theta_1} + l_2 T_2 + \tau_{\theta_2}.$$

- torque around the z -axis (τ_{ψ_0}) is the result of the sum of the torques applied around the z -axis generated by each drone, along with the horizontal forces in the xy -plane, which can be generated by adjusting the UAVs' roll:

$$\tau_{\psi_0} = -\tau_{\phi_1} + \tau_{\phi_2} + \tau_{\psi_1} + \tau_{\psi_2}.$$

In summary, given a reference command vector $\Gamma = [T_0 \ \tau_{\phi_0} \ \tau_{\theta_0} \ \tau_{\psi_0}]^T$, it is possible to solve the linear system to determine the forces and torques that must be applied by the drones.

The proposal of a positioning or a trajectory tracking task for the tandem configuration can be described by the formulation presented in the previous section. First we compute the computed high-level control signals for the drone-cargo set, and then, applying using (23), the inputs send to each drone, which will effectively accomplish the mission.

In term of control, the control law (12) uses the variables related to the pose of the tandem configuration. As well the adaptation law (20) takes the current states of the rigid formation to update the system parameters.

In order to have an initial estimate, when the drones start the transport task, we assume that the tandem parameters are equal to the sum of the parameters of each drone.

IV. RESULTS AND DISCUSSION

To evaluate the effectiveness of the proposed model-based adaptive control strategy, we conducted numerical simulations using a MATLAB-based framework called AuRoRA¹ (Autonomous Robots for Research and Applications) [25]. For this purpose, we adopted the dynamic model of the Tarot 450 quadrotor, an open-hardware UAV developed by the Laboratory for Multirobot Systems at the Czech Technical University in Prague. This model is also implemented in the Gazebo physics simulator and is publicly available through the MRS system repository² [26].

To explore the controller's response under different operating conditions, two simulation scenarios were designed. In the first scenario, a single UAV carries a payload rigidly attached to its body and subsequently releases it mid-flight, requiring the controller to adapt to abrupt changes in system dynamics. In other words, the UAV operates with the payload rigidly coupled to its frame, effectively forming a single dynamic body with increased mass and altered moments of inertia. At 150 seconds, the payload is released—emulating a free-fall event—which imposes a sudden shift in system parameters and serves as a key test of the controller's adaptability.

In the second scenario, the tandem configuration is evaluated to illustrate how the control inputs are distributed across multiple UAVs during cooperative load transport. These simulations were executed sequentially to facilitate a comprehensive visualization of the parameter adaptation process and its role in compensating for variations in flight conditions.

The reference trajectory for both scenarios follows a Bernoulli's lemniscate in three-dimensional space, with altitude variations and a constant heading angle.

¹The interested reader is invited to explore the AuRoRA framework and its main projects at <https://github.com/neroUFV>.

²MRS System repository: <https://github.com/ctu-mrs>

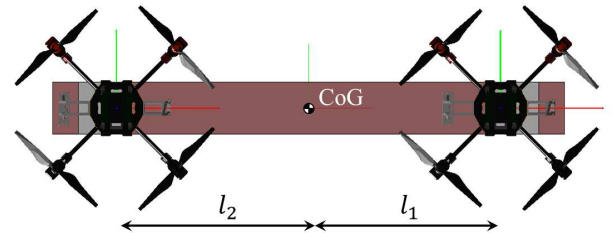


Fig. 2. Distance of the drones related to the center-of-gravity (CoG) of the Tandem configuration.

To better understand the proposal of this work, as well as to facilitate the visualization of the results, the video available at link <https://youtu.be/6xZKyG00sgE> shows the scenarios used to validate the control and adaptation strategy proposed for the cargo transportation task.

A. Adaptive Control for a UAV

In the first experimental scenario, a single UAV is tasked with transporting a payload directly attached to its body, simulating a rigid coupling via a gripper or mechanical interface. This configuration leads to a significant change in the UAV's physical parameters—most notably, an increase in total mass and moments of inertia. These changes directly affect the dynamics and degrade the performance of conventional model-based controllers that rely on nominal parameters.

Initially, the UAV operates under the underactuated control law without parameter adaptation. During this period, the controller fails to fully compensate for the increased payload mass, resulting in visible vertical tracking errors. The thrust produced is insufficient to maintain the reference altitude, as the controller is unaware of the additional load.

At 60 seconds, the adaptive control law is activated. From this point onward, the controller begins to estimate and compensate for the altered dynamics by updating internal representations of the system's mass and inertia. As shown in Figure 3, this adaptation process results in a rapid correction of the vertical trajectory error, bringing the UAV closer to the desired reference within approximately 10 seconds. This confirms the controller's ability to adjust to unknown or time-varying parameters in real-time.

Midway through the experiment (at 150 seconds), the payload is released, generating a sudden disturbance. The UAV, now lighter and with reduced inertia, reacts immediately, causing a sharp upward deviation in altitude. This abrupt transition serves as a critical test of the system's robustness and adaptability. The adaptive controller quickly identifies the shift in dynamics and readjusts the internal parameter estimates accordingly. As observed in the red-shaded region of Figure 3, the control system remains stable throughout the entire adaptation process, with no evidence of overshoot-induced instability or oscillations.

The control error along the z -axis clearly reflects the impact of the sudden vertical disturbance introduced by the payload release at the midpoint of the experiment. In response to this event, the adaptive controller dynamically updates the estimated mass and inertia parameters to reflect the UAV's new physical state. By continuously refining the internal model, the controller adjusts the thrust command accordingly, enabling the UAV to promptly recover and maintain altitude closer to the reference trajectory during the active adaptation phase.

In summary, the results confirm the controller's ability to handle unmodeled disturbances and validate its usefulness in aerial transportation missions involving variable payload conditions.

B. Adaptive Control for the Tandem

In the second scenario, the tandem configuration is employed to validate the proposed adaptive control strategy under cooperative transport conditions. Figure 4 illustrates the time evolution of the UAVs' and payload's position as the system tracks a Bernoulli's lemniscate trajectory at varying altitude.

Three distinct phases can be identified during the simulation. In the first phase, the UAVs initiate the transportation task without any adaptation mechanism active. Due to the added mass and the coupling dynamics of the bar-shaped payload, the system is unable to maintain the desired altitude, resulting in a persistent vertical tracking error. This error highlights the effect of unmodeled payload-induced disturbances, including inertia variation and oscillatory behaviors.

Once the adaptive controller is activated, the second phase begins. The tandem system gradually adapts its internal parameter estimates—mass and moments of inertia—based on the observed dynamics. This adaptation improves altitude tracking as the controller compensates for the previously unaccounted payload characteristics. The increase in altitude seen in this phase indicates that the controller successfully reacts to the disturbance forces, adjusting thrust commands accordingly to stabilize the formation.

The third phase is characterized by oscillations observed in the lateral and longitudinal motion. These oscillations are primarily attributed to the mutual interaction forces between the drones, transmitted through the rigid payload. As each UAV independently attempts to stabilize its share of the load, slight misalignments in control inputs may cause alternating forces that push and pull the structure. This can lead to dynamic coupling and resonance-like behavior, where the UAVs inadvertently excite each other through the bar. While the adaptive control partially mitigates these effects, the lateral fluctuations demonstrate the complex nature of cooperative aerial manipulation, particularly in rigid-body tandem configurations.

V. CONCLUDING REMARKS

This paper presented an adaptive control strategy for cooperative load transportation using quadrotors arranged in a tandem configuration. By modeling the UAVs and the payload as a unified rigid-body system, the proposed framework effectively addresses the complex dynamic interactions inherent in multi-agent aerial transport, including variations in mass distribution, inertia, and inter-agent coupling forces.

The use of an underactuated adaptive control approach enables real-time parameter estimation and compensation, allowing the system to maintain stable and accurate trajectory tracking even under abrupt changes in payload conditions. Numerical experiments demonstrated the robustness of the strategy in two scenarios: (i) a single UAV transporting and releasing a payload, and (ii) a tandem configuration executing a trajectory tracking task while managing dynamic coupling effects. In both cases, the controller adapted to unforeseen disturbances and preserved the system's stability and performance.

The results underscore the potential of adaptive control to enhance the autonomy and robustness of UAV-based cooperative transport systems. Future work will focus on validating the proposed strategy in real-world experiments using physical drones. In particular, we intend to investigate collaborative transport scenarios where a single UAV, upon detecting an overloaded condition, autonomously recruits a second UAV to complete the transport mission cooperatively.

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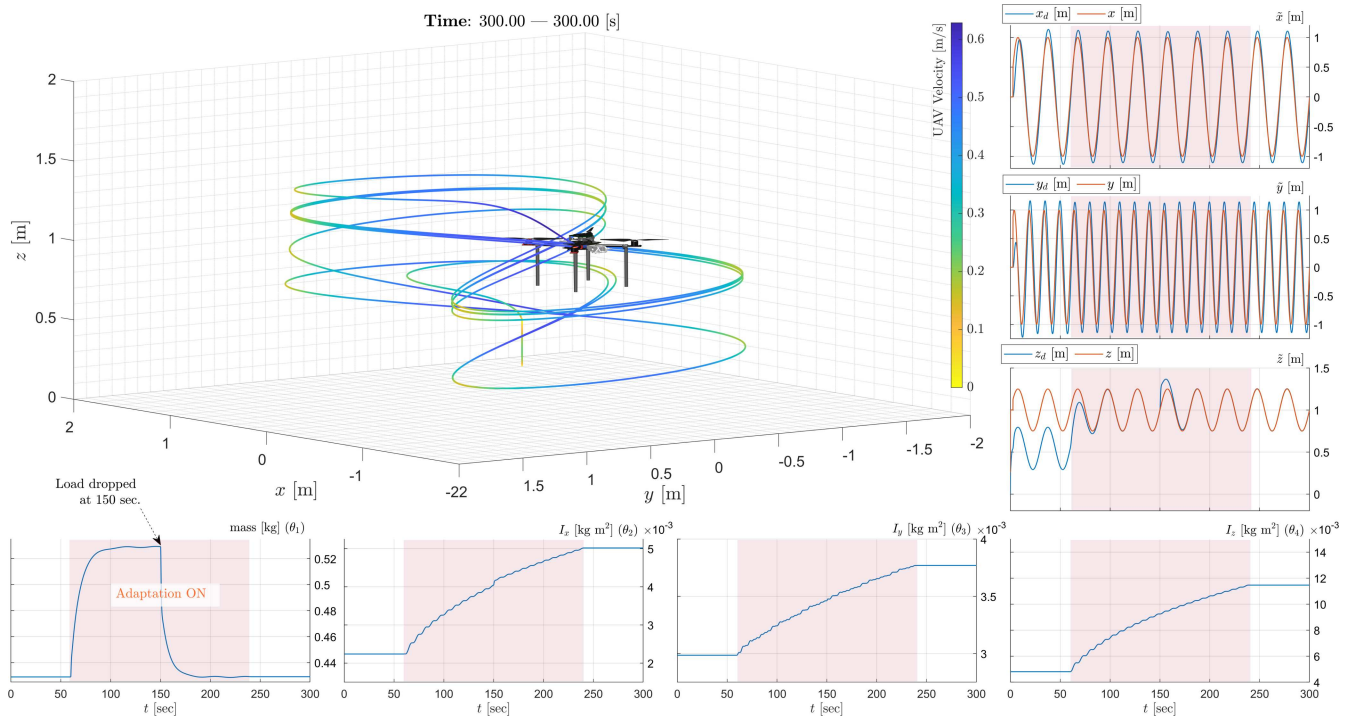


Fig. 3. Position of the UAV and the load during the transportation task. Here the solid orange line represents the displacement of the UAV, while the solid blue line represents the desired reference movement for the system. The adaptation process is applied to the UAV in the interval of 60 to 240 seconds. In this scenario, the load is dropped at 150 seconds and the disturbance caused can be observed in the mass adaptation and in the z -axis coordination error.

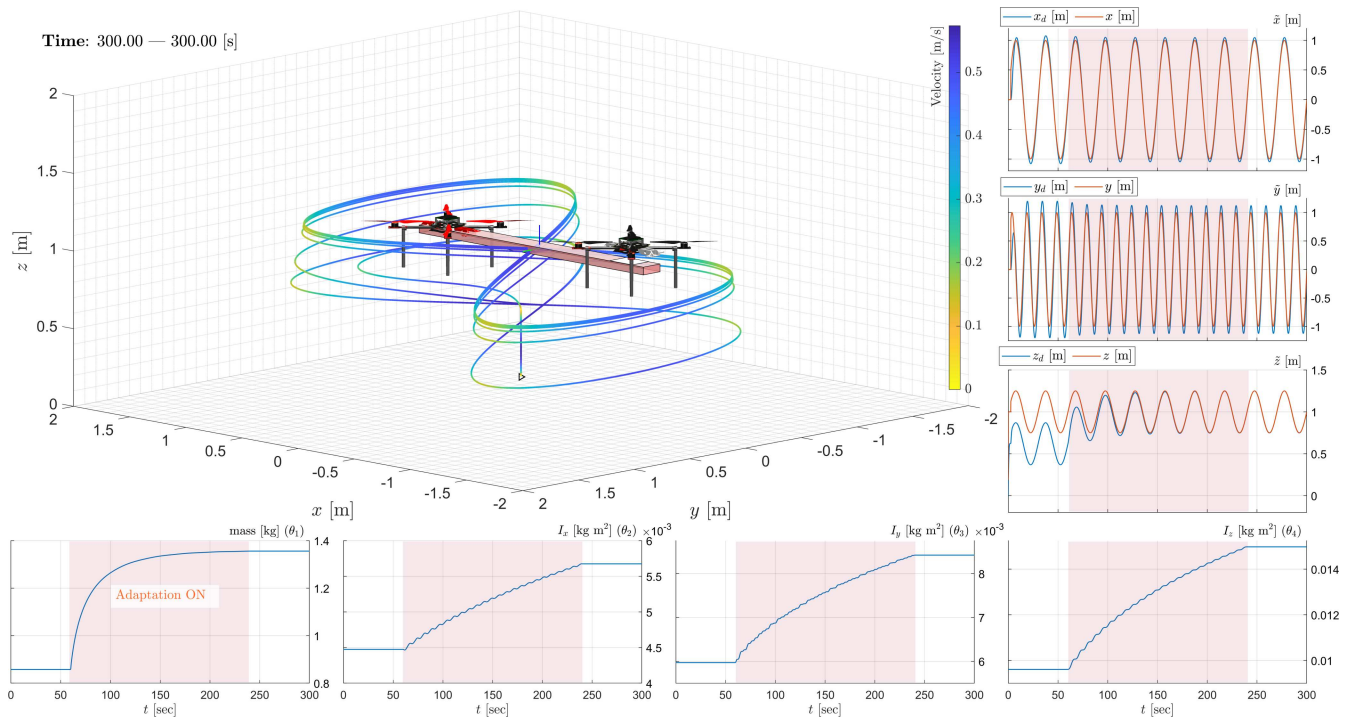


Fig. 4. Adaptation process applied to the tandem configuration during a trajectory tracking task. The bar-shaped payload is requested to follow Bernoulli's lemniscate shape during this experiment. The adaptation underactuated control law is applied during the period time of 60 to 240 seconds. The tandem velocity is depicted in the colorbar.